LightningRL

Policy Optimization

$$\mathcal{L}(\theta) = \mathcal{L}(\theta)_{\text{GRPO}} + \mu \underbrace{\frac{1}{\sum_{\tau \in \mathcal{T}_{\text{succ}}} |\tau|} \sum_{\tau \in \mathcal{T}_{\text{succ}}} \sum_{t=0}^{T-1} \sum_{k \in M_{\tau,t}} \log p_\theta(x_{\tau,t+1}^{(k)} \mid x_{\tau,t}, q)}_{\textbf{Token-level NLL Loss}}$$

Sampling and Reward

Filter

$$\Delta\text{TPF}(q) = \max_j \text{TPF}(\tau_j) - \min_j \text{TPF}(\tau_j) \geq \delta$$

At least one sample is correct

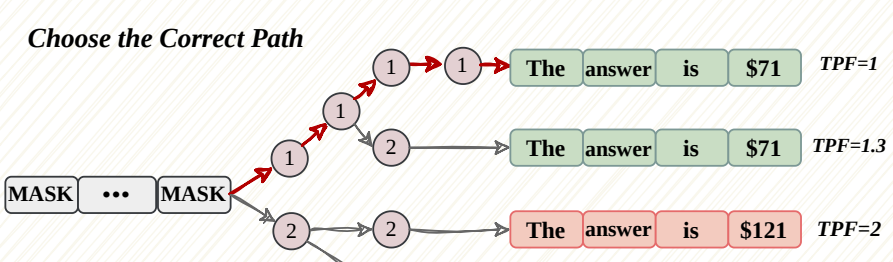
$$\hat{A}^{(i,j)} = \frac{A^{(i,j)} - \text{mean}_{(i',j') \in \mathcal{B}} A^{(i',j')}}{\text{std}_{(i',j') \in \mathcal{B}} A^{(i',j')} + \epsilon}$$

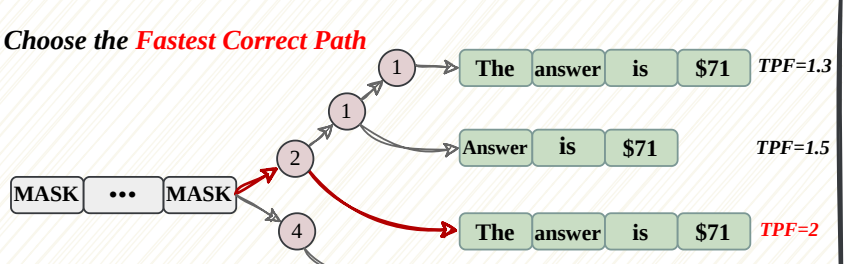
$$R_1^{(i,j)} = \frac{r_1^{(i,j)} - \text{mean}\left\{r_1^{(i,1)}, \ldots, r_1^{(i,G)}\right\}}{\text{std}\left\{r_1^{(i,1)}, \ldots, r_1^{(i,G)}\right\}}, \ \ldots, \ R_n^{(i,j)} = \frac{r_n^{(i,j)} - \text{mean}\left\{r_n^{(i,1)}, \ldots, r_n^{(i,G)}\right\}}{\text{std}\left\{r_n^{(i,1)}, \ldots, r_n^{(i,G)}\right\}} \quad A^{(i,j)} = \sum_{k=1}^{n} R_k^{(i,j)}.$$

***Prompt*: Ali had $21. Leila gave him half of $100. How much does Ali have now?**

Before Training

Choose the Correct Path



After Training

Choose the Fastest Correct Path